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EXPERIMENT-2

AIM :

Live Forensics Case Investigation using Autopsy . Here is the procedure and expected result for Experiment 2 (Live Forensics Case Investigation using Autopsy):

REQUIREMENTS:

1. Computer running Windows, Linux, or another supported operating system.
2. Autopsy forensic investigation platform.
3. A sample disk image or authorized test evidence.
4. Sufficient disk space for the case database and extracted artifacts.

PROCEDURE:

Step 1: Download and Install Autopsy

Visit the official website: Autopsy Download.
Download the appropriate installer.
Run the installer and complete the installation.
Launch Autopsy from the desktop or Start menu.

Step 2: Create a New Case

Open Autopsy and select New Case.
Enter a suitable case name, such as Digital_Forensics_Lab.
Select a base directory with sufficient storage space.
Enter the case number and examiner details if prompted.
Click Finish to create the case.

Step 3: Add a Data Source

Select Add Data Source in the case interface.
Choose the appropriate source type, such as a disk image or logical files.
Browse to the authorized evidence file.
Verify that the correct source has been selected.
Click Next to configure processing options.

Step 4: Configure Ingest Modules

Select the relevant ingest modules available in your Autopsy version.
Enable modules for file identification, metadata extraction, operating system artifacts, and web activity as applicable.
Enable hash calculation or known-file identification if required.
Avoid selecting every module blindly; use modules relevant to the evidence and available system resources.
Click Next or Finish, depending on the version.

Step 5: Start Evidence Processing

Click Finish to begin processing the data source.
Autopsy analyzes the selected evidence and extracts supported artifacts.
Wait until processing completes or check the ingest progress.
Review any warnings or errors displayed.

Step 6: Examine Investigation Results

Expand the relevant sections in the left navigation panel and inspect the extracted artifacts.

Section	Information examined
Devices Attached	Connected-device artifacts, if available
EXIF Metadata	Image details, camera information, and timestamps
Installed Programs	Installed application records
Operating System Information	OS version and system details
Operating System User Accounts	User account information
Recent Documents	Recently accessed document artifacts
Web History and Bookmarks	Browsing activity
Web Downloads and Cookies	Download records and browser cookies
Web Search	Recovered search-related artifacts
Email Addresses	Email addresses identified during pro cessing

Step 7: Examine Timestamps and Metadata

- Select an image, document, or other relevant artifact.
- Open its file metadata or details panel.
- Record available creation, modification, and access timestamps.
- Note the displayed time zone and timestamp format where available.
- Capture the filename, relevant metadata, and associated time information.

Important: File timestamps may be missing, altered, or interpreted differently across time zones. Record the values exactly as displayed and avoid assuming that every timestamp represents the actual time of an event.

Step 8: Prepare the Investigation Report

- Review the relevant findings.
- Record the evidence source and case number.
- Include the important artifact details and timestamps.
- Save screenshots as supporting evidence.
- Export a report using the reporting options available in your Autopsy version.

S. No.	Activity	Observation
1	Case creation	Case created successfully
2	Data source addition	Evidence source selected
3	Ingest processing	Artifacts processed
4	Metadata examination	File details and timestamps inspected
5	Artifact analysis	System and browser artifacts reviewed
6	Report preparation	Findings documented

OUTPUT:

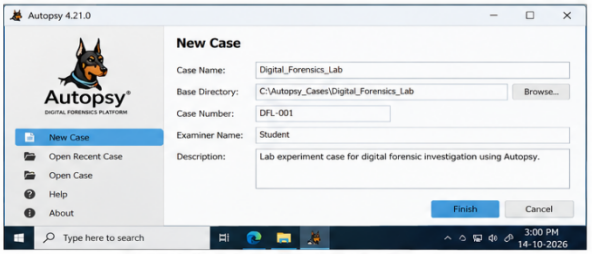


Figure 1: Creating a New Case in Autopsy

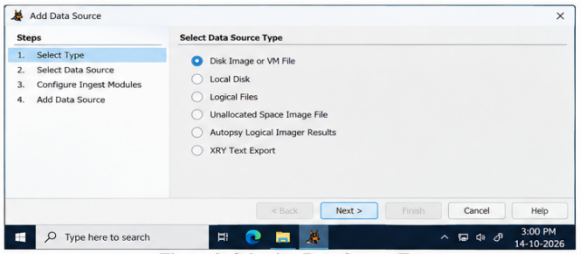


Figure 2: Selecting Data Source Type

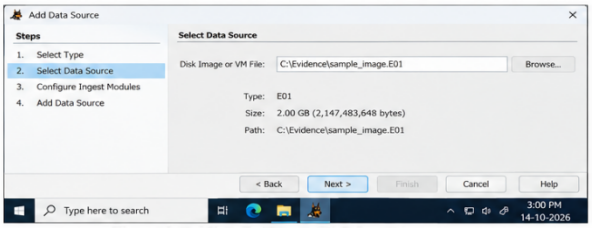


Figure 3: Adding Evidence (Disk Image)

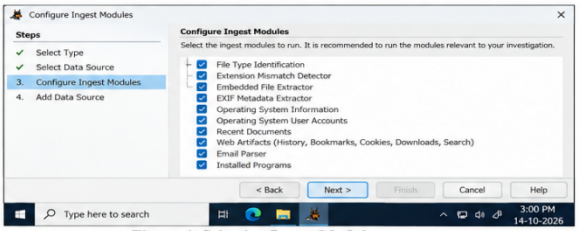


Figure 4: Selecting Ingest Modules

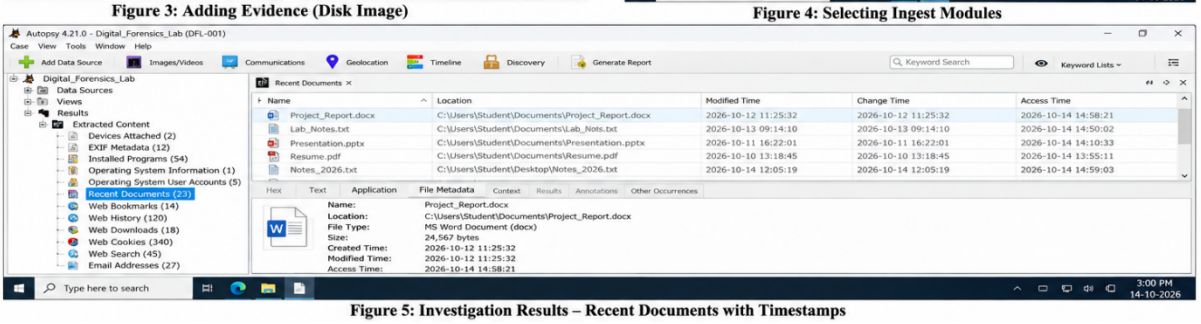


Figure 5: Investigation Results – Recent Documents with Timestamps

RESULT:

The digital forensic investigation procedure using Autopsy was studied. A case was created, an evidence source was added, relevant ingest modules were configured, and available system information, metadata, timestamps, user accounts, and browser artifacts were examined.